

TRACK 4 — PRODUCT & SERVICE INNOVATION

Apply design thinking and engineering principles to build tangible products, digital platforms, and optimised services that enhance quality of life.

Challenge 4.1 — Volunteer–NGO Matching Platform

Domain / Track: Product Innovation

Background & Context Indian college students consistently report wanting to volunteer but struggle to find verified, skill-matched, time-feasible opportunities. NGOs simultaneously report receiving floods of unskilled applicants and lacking the bandwidth to vet them. Existing platforms (iVolunteer, Volunteers For India) exist but suffer from low NGO verification depth and weak skill-matching.

Problem Statement Architect a digital platform that intelligently matches student volunteers to verified NGO projects based on skills, availability, location, and cause area — with built-in milestone tracking, impact logging, and a two-way feedback loop.

Why This Matters Better matching unlocks the latent volunteer capacity of India's 4+ crore college students. A platform that converts even 1% more would inject millions of skilled hours into the social sector.

Project Objectives

- Define 3 user personas (student, NGO coordinator, college NSS unit head).
- Build a clickable high-fidelity prototype OR a functional MVP.
- Design a matching algorithm (rules-based is acceptable for v1).
- Document a go-to-market plan for 3 pilot colleges.

Key Questions to Explore

- What is the cold-start problem — chicken (volunteers) or egg (NGOs)?
- How do you verify NGO legitimacy at scale (Darpan ID, 80G, FCRA)?
- How do you prevent ghosting on both sides?

Suggested Approach

- Use Lean UX: persona → JTBD → wireframes → prototype.
- v1 matching: rules-based (cause + skill + location + hours).
- v2 (stretch): vector embeddings of project + volunteer profile.

Recommended Tools & Resources

- Stack: Next.js / React Native, Firebase / Supabase, Tailwind.
- Design: Figma, Maze for user testing.
- Verification: NGO Darpan portal API, MCA records.

Expected Deliverables

- Clickable Figma prototype OR functional MVP.
- System architecture document.
- Go-to-market and onboarding plan.

Success Metrics

- ≥ 10 user tests with SUS (System Usability Scale) score > 70 .
- At least 3 NGOs verbally commit to listing.
- Matching algorithm tested on a synthetic dataset of ≥ 100 profiles.

Potential Impact A free, NSS-backed alternative that could become the default volunteering pipeline across IITs, NITs, and central universities.

Challenge 4.2 — Community Issue Reporting & Tracking System

Domain / Track: Product Innovation

Background & Context Existing civic platforms — Swachhata App, MCGM 1916, Bengaluru's BBMP Sahaaya — show that citizen reporting works, but suffer from low transparency, no escalation when complaints rot, and limited multilingual support. A 2023 Janaagraha audit found median resolution time on city portals exceeded 14 days, and citizens rarely received closure proof.

Problem Statement Build a citizen-first portal/app for reporting sanitation, infrastructure, encroachment, and public-safety issues — with photo/geo-tag capture, real-time tracking, automated SLA escalation, anonymous-submission options, and a public dashboard showing ward-level resolution performance.

Why This Matters Transparent public dashboards have been shown (Janaagraha, Praja) to nudge municipal officers toward faster resolution. The accountability loop is the product, not the form.

Project Objectives

- Build a functional prototype (web or mobile) with citizen, admin, and public-dashboard views.
- Design a clear escalation policy.
- Address data privacy (consent, anonymisation).

Key Questions to Explore

- How do you handle false or duplicate reports?
- What is the right SLA per issue type — pothole vs. open manhole vs. garbage?
- How does the system avoid becoming a tool for harassment of vendors / minorities?

Suggested Approach

- Build on React Native (mobile) or Next.js (PWA).
- Use Mapbox / Google Maps for geo-tagging.
- Design clear data-flow diagrams and a privacy policy aligned with DPDP Act, 2023.

Recommended Tools & Resources

- Reference: Bhuvan citizen-services platform, Swachhata SBM-U App, FixMyStreet (UK open-source).
- Stack: Next.js, Supabase, Mapbox, Cloudinary.
- DPDP Act 2023 compliance checklist.

Expected Deliverables

- Functional prototype with all 3 views.
- Admin dashboard mockup.
- Data-privacy + escalation policy document.

Success Metrics

- End-to-end report-to-resolution flow demonstrable.
- $\geq 8/10$ usability rating from 10 test users (preferably from diverse age groups).
- Clear DPDP compliance walkthrough.

Potential Impact A municipality-ready open-source civic tool — particularly valuable for Tier-2/3 cities that cannot afford bespoke platforms.

Challenge 4.3 — Service Solutions for Accessibility & Inclusion

Domain / Track: Product Innovation

Background & Context India ratified the UNCRPD in 2007 and passed the RPwD Act in 2016, yet implementation lags. Public buses, ration shops, polling booths, and even most college campuses are functionally inaccessible. Open-source assistive tech (e.g., the Mukti Bionics arm, Bolt smart cane) has shown that ₹2,000–₹10,000 interventions can replace ₹50,000+ imports.

Problem Statement Pick ONE — either (a) design a low-cost open-source assistive technology for a specific disability use case, OR (b) redesign one specific public service workflow (e.g., pension disbursement for the elderly, polling-station experience for wheelchair users) to be radically inclusive.

Why This Matters Accessibility is not a niche — it benefits the elderly (15 crore Indians by 2030), pregnant women, the temporarily injured, and linguistic minorities.

Project Objectives

- Define the user with primary interviews (at least 5 users with the relevant lived experience).
- Build a working prototype OR a publishable design specification.
- Cost the solution end-to-end.

Key Questions to Explore

- What is the cheapest version that still preserves user dignity?
- What support ecosystem (repair, training, replacement) does the user need?
- How is it different from existing solutions and why hasn't anyone built it yet?

Suggested Approach

- Use inclusive-design tools from Microsoft's Inclusive Design Toolkit.
- Hardware: 3D-printed parts, Arduino/ESP32, off-the-shelf modules.
- Service redesign: Service Blueprint + journey map.

Recommended Tools & Resources

- RPwD Act 2016, WCAG 2.2 guidelines for digital.
- Communities: Enable India, Vision Empower, Mitti Cafe.
- Tools: TinkerCAD, Fusion 360, Figma.

Expected Deliverables

- Prototype OR detailed design specification with BOM.
- Accessibility compliance checklist.
- Cost estimate for manufacture or deployment.

Success Metrics

- Validated by ≥ 5 users with the relevant lived experience.
- Per-unit cost $\leq 25\%$ of the closest commercial alternative.

Potential Impact A reference design that DPOs (Disabled People's Organisations) and panchayats can manufacture locally.

